

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Document Version: 1.0.0 Architecture: Enterprise Service Management & Flow Orchestration

1. INTRODUCTION

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team Members & Roles

- **Lead Systems Engineer:** Workflow Architecture
- **ServiceNow Developer:** Flow Designer & Scripts
- **Asset Manager Lead:** CMDB & Inventory Integration

2. PROJECT OVERVIEW

Purpose

The goal of this project is to build an automated procurement portal and backend execution engine that streamlines employee hardware orders. By replacing manual ticketing with an automated request orchestration system, the application handles order validation, approval hierarchy routing, inventory checking, and vendor fulfillment triggers.

Key Features

- **Self-Service Service Portal:** Dynamic catalog form for hardware configuration selection.
- **Dynamic Approval Routing:** Automated manager approval triggers with multi-tier escalation logic.
- **Automated Stock Management:** Real-time CMDB inventory lookup and auto-reservation.
- **Vendor API IntegrationHub:** Automated REST dispatch for third-party supplier orders when local inventory is low.

3. ARCHITECTURE

3.1 Frontend

Built using ServiceNow Service Portal (AngularJS / HTML5 / Bootstrap). Provides responsive request forms, variable validation, and real-time order status tracking widgets.

3.2 Backend

ServiceNow Flow Designer engine executing server-side JavaScript (ES6) actions, decision tables, sub-flows, and IntegrationHub REST steps.

3.3 Database

Relational ServiceNow CMDB schema storing Request Items (sc_req_item), Catalog Tasks (sc_task), and Hardware Assets (alm_hardware).

4. SETUP INSTRUCTIONS

Prerequisites

- ServiceNow Instance (Utah / Vancouver / Washington release).
- IntegrationHub Enterprise Plugin enabled.
- Admin or flow_designer & catalog_admin roles assigned.

Installation Steps

1. Clone the project update set repository or download the XML payload.
2. Navigate to **System Update Sets > Retrieved Update Sets** in ServiceNow.
3. Import and preview sys_remote_update_set_procurement_flow_v1.xml.
4. Resolve any target conflicts and click **Commit Update Set**.
5. Publish the catalog item **Standard Employee Laptop** to the Service Portal.

5. FOLDER STRUCTURE

Client / Portal Assets

```
/sp_portal/catalog_items/ ├── standard_laptop_form.xml ├── client_scripts/ | ├── validate_cost_center.js | └── dynamic_model_filter.js └── ui_policies/ └── toggle_spec_details.xml
```

Server / Flow Components

```
/sys_hub_flow/procurement_engine/ ├── main_flow.xml ├── actions/ | ├── check_cmdb_stock.js | └── dispatch_vendor_po.js └── subflows/ └── manager_escalation_subflow.xml
```

6. RUNNING THE APPLICATION

To launch and execute the end-to-end flow locally or in a dev environment:

- **User Portal:** Access `https://<instance>.service-now.com/sp?id=sc_cat_item&sys_id=laptop_cat_id` and submit an order.
- **Flow Operations Center:** Access **Flow Designer > Executions** to view live trigger states and context steps.

7. API DOCUMENTATION

Endpoint	Method	Parameters	Response Example
/api/now/table/sc_req_item	POST	cat_item, requested_for, cost_center	{"result": {"sys_id": "47a9f...", "number": "RITM001029"}}
/api/v1/vendor/laptop-order	POST	model_sku, delivery_address, po_number	{"status": "ACKNOWLEDGED", "order_id": "V-89102"}

8. AUTHENTICATION

User authentication relies on corporate Single Sign-On (SSO / SAML 2.0). External vendor API calls generated by Flow Designer IntegrationHub steps utilize OAuth 2.0 Client Credentials flow with encrypted secret storage in ServiceNow Credentials Vault.

9. USER INTERFACE & DEMO SCREENSHOTS

The UI features a clean multi-step request catalog form, real-time approval status badges, and an automated fulfillment timeline visible to both the employee and the provisioning team.

10. TESTING STRATEGY

- **Automated Test Framework (ATF):** Automated suite runs on catalog submission and variable mapping validation.
- **Flow Engine Simulation:** Execution context testing across threshold edge-cases (e.g., requests > \$1,500 vs. standard requests).

11. KNOWN ISSUES & FUTURE ENHANCEMENTS

12. Known Issues

- Vendor API timeout on peak days can cause flow pauses (handled by retry policy).
- Manual address overrides require fallback to IT Asset queue.

13. Future Enhancements

- Predictive hardware ordering using machine learning on onboarding trends.
- Integration with RFID/BLE asset tracking upon box delivery.